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This project examines whether an inverted Treasury yield curve is a reliable early warning sign of a U.S. recession and weaker stock market returns, and how far in advance it signals trouble. Normally, long-term government bonds pay higher interest rates than short-term bonds. When the 2-year rate rises above the 10-year rate, the yield curve is considered "inverted," which many investors view as a recession signal. Using data from 1976 to today, we will measure how often inversions were followed by a recession, how much lead time they provided, and how stock returns, unemployment and inflation behaved in the year that followed. We will also assess whether the signal still holds after the record-long 2022–2024 inversion.

This question matters because an early warning gives management time to adjust hiring, capital spending, and borrowing before demand slows. Stock returns are also important to companies because they affect pension funding, investment values, and the timing of stock buybacks or raising new capital. Unemployment and inflation matter because they are closely linked. Inversions often happen when the Federal Reserve raises rates to fight high inflation, and the slowdown that follows usually brings rising unemployment and cooling prices. From an accounting perspective, downturns affect key estimates such as loan loss reserves, impairment testing and going-concern judgments. Prior research found that the yield curve was the best single predictor of recessions more than one quarter ahead. However, the 2022–2024 inversion was the longest on record, has not yet been followed by a recession and saw strong stock gains, which raises the question of whether the signal is broken or simply delayed. We expect inversions to be followed by a higher chance of recession within 12 months, warning times of roughly 6 to 24 months, rising unemployment, slowing inflation, and below-average stock returns.

We will use free public data from the Federal Reserve Economic Data (FRED) website, including the 10-year minus 2-year Treasury spread, the unemployment rate, the Consumer Price Index (CPI), an official recession indicator and the federal funds rate. For stock returns, we will use Robert Shiller's publicly available monthly S&P 500 data, which covers our full time period. All data work will be done in Python. We will convert daily interest rates to monthly averages, create new variables such as an inversion flag, a "recession within twelve months" flag, the yearly inflation rate from CPI, and the change in unemployment and S&P 500 return over the following 12 months, and then join everything into one monthly dataset. Our analysis will compare recession rates, stock returns, unemployment and inflation after inverted versus normal periods, measure the lead time for each inversion, and use regression to estimate how the spread relates to recession probability and future stock returns. We will check our results against FRED's own figures, official recession dates, and published S&P 500 returns, and present our findings in a few clear charts.